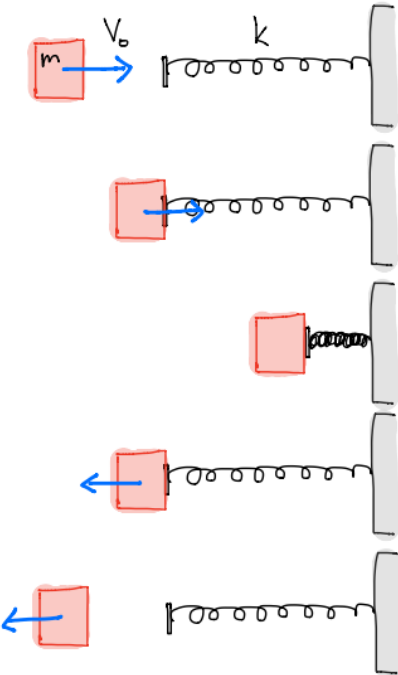


Week 1

1. A 500 g block slides along a frictionless surface at a speed of 0.35 m/s. It runs into a horizontal(水平的) massless(无质量的) spring(弹簧) with spring constant(劲度系数) 50 N/m that extends(延伸) outward from(类似: push forward to) a wall. It compresses(压缩) the spring, and then is pushed back in the opposite direction(反方向的) by the spring, eventually losing contact with(脱离) the spring.

(a) How long (多长时间) does the block remain in contact with the spring? (It might help to draw a diagram of the system at several points in time.)

undergo(经历)



While the mass is touching the spring, it undergoes half a period of SHM.

$$\begin{aligned} \therefore \text{time in contact} &= \frac{1}{2}T \\ &= \frac{1}{2} \cdot 2\pi \sqrt{\frac{m}{k}} \\ &= \pi \sqrt{\frac{0.50 \text{ kg}}{50 \text{ N/m}}} \\ &= \boxed{0.31 \text{ s}} \end{aligned}$$

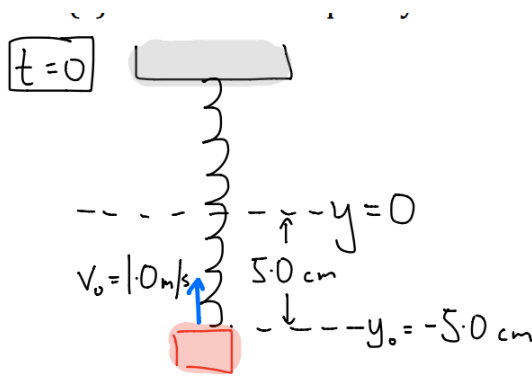
(b) How would your answer to part (a) change if the block's initial speed were doubled?

The period only depends on the mass m and the spring stiffness k . So changing the initial speed would not change the answer to part (a)...

UNLESS the spring gets compressed so much that $F_s = -kx$ is not true anymore!

2. A 5.0 kg block hangs from (悬挂在) a spring with spring constant 2000 N/m. The block is pulled down (拉长) 5.0 cm from the equilibrium position and given an initial velocity of 1.0 m/s back toward equilibrium (平衡位置).

(a) What is the frequency of the motion?



$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{2000 \text{ N/m}}{5.0 \text{ kg}}} = \boxed{3.2 \text{ Hz}}$$

(b) What is the amplitude of the motion?

Use **ENERGY CONSERVATION**:

$$E_{\text{tot}} = K_i + U_i = U_{\text{max stretch}}$$

$$\frac{1}{2}mv_0^2 + \frac{1}{2}ky_0^2 = \frac{1}{2}kA^2$$

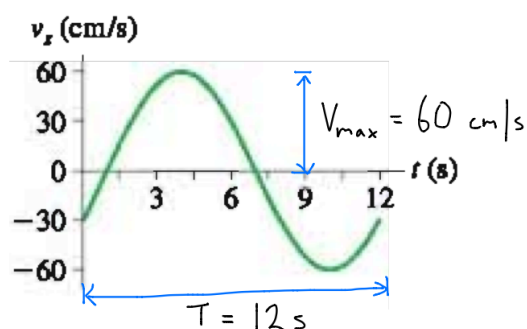
$$\rightarrow A = \sqrt{\frac{m}{k}v_0^2 + y_0^2} = \sqrt{\frac{5.0}{2000}(1.0 \text{ m/s})^2 + (-5.0 \times 10^{-2} \text{ m})^2}$$

$$= 0.071 \text{ m or } \boxed{7.1 \text{ cm}}$$

(c) What is the total mechanical-energy (机械能=K+U) of the motion?

$$\begin{aligned} \text{From (b): } E_{\text{total}} &= \frac{1}{2} k A^2 \\ &= \frac{1}{2} (2000 \text{ N/m}) (0.071 \text{ m})^2 \\ &= \boxed{5.0 \text{ J}} \end{aligned}$$

3. The figure below is the velocity-versus(与)-time (not position-versus-time!) graph of a particle in simple harmonic motion.



(a) What is the amplitude of the oscillation?

For SHM, we know

$$\begin{aligned} x(t) &= A \cos(\omega t + \phi_0) \\ \rightarrow v_x(t) &= \frac{dx}{dt} = -\omega A \sin(\omega t + \phi_0) \end{aligned}$$

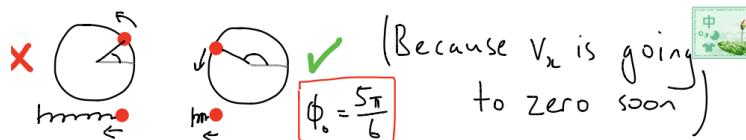
$$\begin{aligned} \therefore v_{\max} &= \omega A = \frac{2\pi}{T} A \\ \hookrightarrow A &= \frac{T v_{\max}}{2\pi} = \frac{(12 \text{ s})(0.6 \text{ m/s})}{2\pi} \\ &= \boxed{1.15 \text{ m}} \end{aligned}$$

(b) What is the phase constant?

Initial condition:

$$v_x(0) = -v_{\max} \sin \phi_0 = -30 \text{ cm/s}$$

$$\therefore \phi_0 = \sin^{-1}\left(\frac{30}{60}\right) = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6} \text{ or } \frac{5\pi}{6} \text{ radians... which one?}$$



Another way: $a_x = \frac{dv_x}{dt} = -\omega^2 A \cos(\omega t + \phi_0)$
 $a_x(t=0) = -\omega^2 A \cos \phi_0 > 0$
 $\therefore \cos \phi_0 < 0$
 $\therefore \phi_0$ must be $\frac{5\pi}{6}$

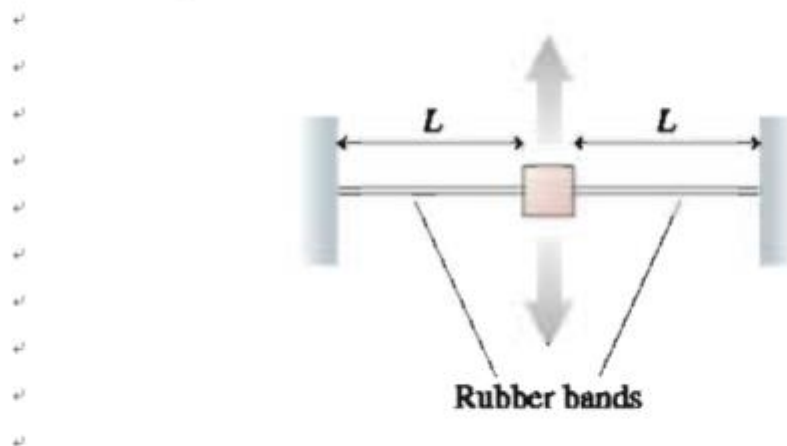
(c) What is the position at $t=0$ s?

$$\begin{aligned}
 x(0) &= A \cos \phi_0 \\
 &= (1.15 \text{ m}) \cos\left(\frac{5\pi}{6}\right) \\
 &= -1.00 \text{ m}
 \end{aligned}$$

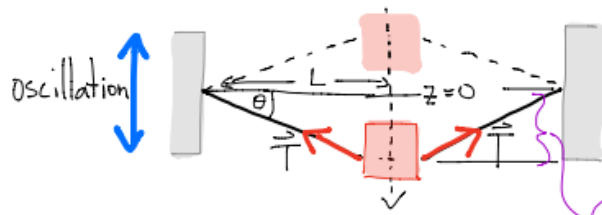
Week 2

1.

The figure below is a top view of an object of mass m connected between two stretched rubber bands (橡皮筋) of length L . The object rests on a frictionless surface. At equilibrium, the tension in each rubber band is T .



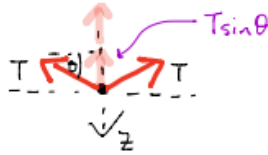
Find an expression (求一个公式) for the frequency of oscillations *perpendicular to* (垂直于) the rubber bands. You can assume that the magnitude of the tension in the rubber bands does not change as the mass oscillates.



SYSTEM: mass

SURROUNDINGS: two rubber bands

FREE-BODY
DIAGRAM:



$$z = L \tan \theta$$

NEWTON'S 2nd LAW:

$$\frac{d\vec{p}}{dt} = \vec{F}_{\text{net}}$$

$$\boxed{z} \quad m \frac{d^2 z}{dt^2} = -2T \sin \theta$$

$$\frac{d^2 z}{dt^2} = -\frac{2T}{m} \sin \theta$$

Assume small θ ; then $\sin \theta \approx \tan \theta = \frac{z}{L}$

$$\therefore \frac{d^2 z}{dt^2} = -\frac{2T}{mL} z$$

This is the differential equation for SHM, with

$$\omega = \sqrt{\frac{2T}{mL}}$$

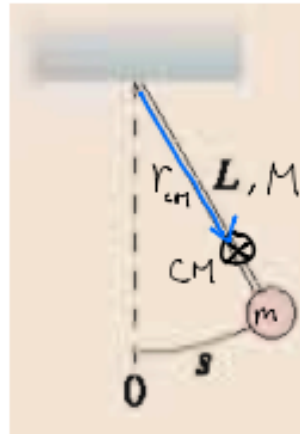
$$\text{i.e. } f \equiv \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{2T}{mL}}$$

2. A 15-cm-long, 200 g rod (杆) is pivoted at one end. A 20 g ball of clay (黏土) is stuck on the other end. What is the period if the rod and clay swing as a pendulum?
inertia(惯性), pendulum(摆)

Center of mass:

$$\begin{aligned} r_{cm} &= \frac{Mr_{rod} + mr_{clay}}{M+m} \\ &= \frac{M(\frac{L}{2}) + mL}{M+m} \\ &= \left(\frac{\frac{M}{2} + m}{M+m}\right)L \end{aligned}$$

(assuming small θ !)



Moment of inertia:

Model the clay ball as a point mass.

$$\begin{aligned} I &= I_{rod} + I_{clay} \\ &= \frac{1}{3}ML^2 + mL^2 \\ &= \left(\frac{M}{3} + m\right)L^2 \end{aligned}$$

For any physical pendulum, we know

$$\omega = \sqrt{\frac{r_{cm} m_{tot} g}{I}}$$

$$\rightarrow T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{r_{cm} m_{tot} g}}$$

$$= 2\pi \sqrt{\frac{\left(\frac{M}{3} + m\right)L^2}{\left(\frac{\frac{M}{2} + m}{M+m}\right)L \cdot \cancel{(M+m)}g}}$$

$$= 2\pi \sqrt{\left(\frac{\frac{M}{3} + m}{\frac{M}{2} + m}\right) \frac{L}{g}}$$

$$= 2\pi \sqrt{\left(\frac{\frac{200}{3} + 20}{\frac{200}{2} + 20}\right) \frac{0.15\text{ m}}{9.8\text{ m/s}^2}} = \boxed{0.66\text{ s}}$$

3. It is said (据说) that Galileo (伽里略) discovered a basic principle of the pendulum that the period is independent of the amplitude – by using his pulse (脉搏) to time the period of swinging lamps (吊灯) in the cathedral (大教堂) as they swayed (摆动) in the breeze (微风). Suppose that one oscillation (振动) of a swinging lamp takes 5.5 s.

(a) How long is the lamp chain?

simple pendulum (单摆)



Model the lamp as a simple pendulum.

Then $\omega = \sqrt{\frac{g}{L}}$

$$\rightarrow T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{L}{g}}$$

$$\rightarrow L = g \left(\frac{T}{2\pi} \right)^2 = (9.8 \text{ m/s}^2) \left(\frac{5.5 \text{ s}}{2\pi} \right)^2 = \boxed{7.5 \text{ m}}$$

(b) What maximum speed does the lamp have if its maximum angle from the vertical is 3.0° ?

vertical is 3.0° ?

$$s(t) = s_{\max} \cos(\omega t + \phi_0) \quad (\text{SHM})$$

$$\rightarrow v_s(t) = \frac{ds}{dt} = -\omega s_{\max} \sin(\omega t + \phi_0)$$

$$\theta_{\max} = 3^\circ = 3 \times \frac{\pi}{180} \text{ radians} = 0.052 \text{ radians}$$

$$V_{\max} = \omega s_{\max} \quad (\text{since the maximum of } |\sin(\omega t + \phi_0)| = 1)$$

$$= \omega L \theta_{\max} \quad (\text{since } s = L\theta \text{ for circular motion})$$

$$= \sqrt{\frac{g}{L}} \cdot L \theta_{\max} = \sqrt{gL} \theta_{\max} = \sqrt{(9.8 \text{ m/s}^2)(7.5 \text{ m})} \cdot (0.052) = \boxed{0.45 \text{ m/s}}$$

Week 3

1. A wave on a string is described by $D(x,t) = (2.0\text{cm}) \times \sin[(12.57\text{rad/m})x - (638\text{rad/s})t]$, where x is in meters(in+单位) and t is in seconds. The linear density(线密度) of the string is 5.00 g/m . What are displacement 位移,

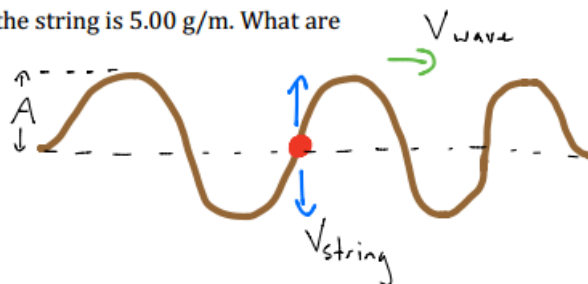
and t is in seconds. The linear density of the string is 5.00 g/m . What are

(a) the string tension?

$$\bullet v_{\text{wave}} = \lambda f = \left(\frac{\lambda}{2\pi}\right)(2\pi f) = \frac{\omega}{k}$$

$$\bullet v_{\text{wave}} = \sqrt{\frac{F_T}{\mu}}$$

$$\sqrt{\frac{F_T}{\mu}} = \frac{\omega}{k} \rightarrow F_T = \mu \left(\frac{\omega}{k}\right)^2$$



(b) the maximum displacement of a point on the string? $= (5 \times 10^{-3}) \left(\frac{638}{12.57}\right)^2 = 12.9\text{ N}$

$$D_{\text{max}} \equiv A = 2.00 \times 10^{-2}\text{ m}$$

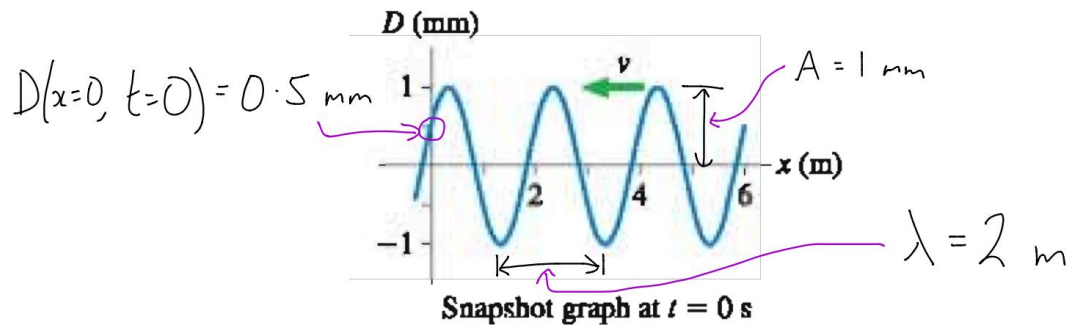
(c) the maximum speed of a point on the string? [CAREFUL: This is *not* the same thing as the wave speed!]

$$v_{\text{string}} \equiv \frac{dD}{dt} = -\underline{\omega A} \cos(kx - \omega t)$$

$$\rightarrow v_{\text{string, max}} = \omega A = (638 \frac{\text{rad}}{\text{s}})(2.00 \times 10^{-2}\text{ m})$$

$$= 12.8 \text{ m/s}$$

2. The figure below is a snapshot(快照) graph at $t = 0$ s of a 5.0 Hz wave traveling to the left.



(a) What is the wave speed?

$$v = \lambda f = (2 \text{ m})(5 \text{ Hz}) = 10 \text{ m/s}$$

(b) What is the phase constant of the wave?

$$D(x=0, t=0) = (1 \text{ mm}) \sin(\phi_0) = 0.5 \text{ mm}$$

$$\phi_0 = \sin^{-1}(0.5) = \frac{\pi}{6} \text{ rad} \text{ or } \frac{5\pi}{6} \text{ rad}$$

$\frac{dD}{dx} > 0$ $\frac{dD}{dx} < 0$

(c) Write the displacement equation ($D(x,t)$) of the wave.

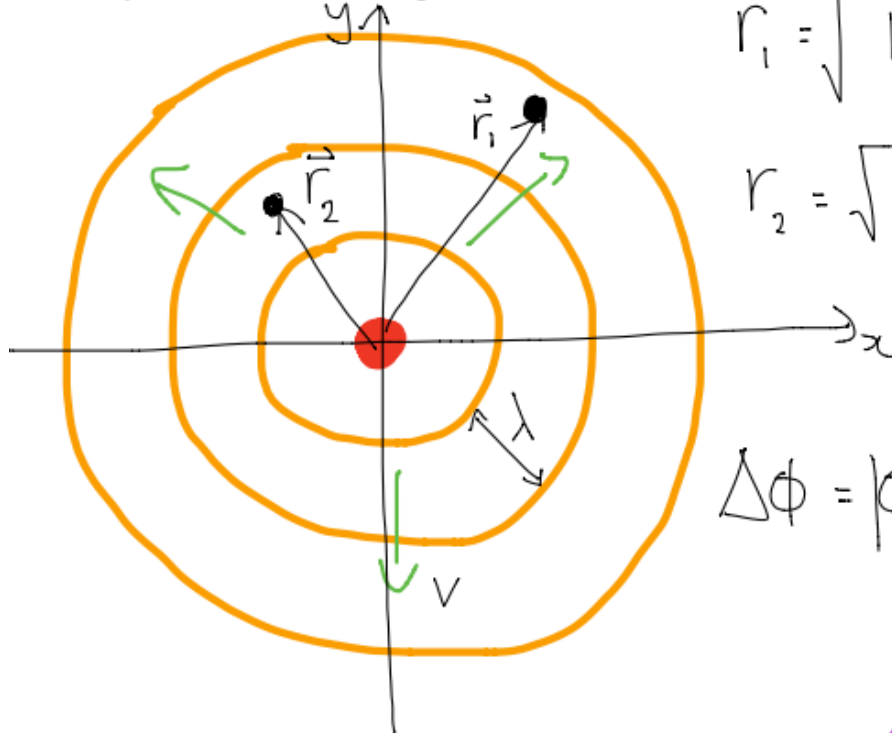
(c) Write the displacement(位移) equation $D(x, t)$ of the wave.

$$D(x, t) = (1 \text{ mm}) \sin \left[\left(\pi \text{ rad/m} \right) x \overset{\text{moving left}}{\downarrow} \left(10 \pi \text{ rad/s} \right) t + \frac{\pi}{6} \text{ rad} \right]$$

$\uparrow A$ $\uparrow k = \frac{2\pi}{\lambda}$ $\uparrow \omega = 2\pi f$ $\uparrow \phi_0$

3. A spherical(球形的) sound source at the origin emits(发出) a sound wave with frequency 13,100 Hz and wave speed 346 m/s. What is the phase difference between the two locations $\langle 1.00, 3.00, 2.00 \rangle$ cm and $\langle -1.00, 1.50, 2.50 \rangle$ cm? Give your answer in both degrees and radians.

your answer in both degrees and radians.



$$r_1 = \sqrt{1^2 + 3^2 + 2^2} = 3.74 \text{ cm}$$

$$r_2 = \sqrt{(-1)^2 + 1.5^2 + 2.5^2} = 3.08 \text{ cm}$$

$$\Delta\phi = |\phi_2 - \phi_1| = 2\pi \frac{|r_2 - r_1|}{\lambda}$$

$$= 2\pi \frac{6.6 \times 10^{-2} \text{ m}}{2.64 \times 10^{-2} \text{ m}}$$

$$= \boxed{\frac{\pi}{2} \text{ rad}}$$

$$= \boxed{90^\circ}$$

$$v = \lambda f$$

$$\Rightarrow \lambda = \frac{v}{f} = \frac{346}{13100} = 2.64 \times 10^{-2} \text{ m}$$

Week 4

1. Two train whistles(发出汽笛声), A and B , each have a frequency of 392 Hz. A is stationary(静止的) and B is moving toward the right (away from A) at a speed of 35.0m/s. A listener is between the two whistles and is moving toward the right with a speed of 15.0m/s (Figure below). No wind is blowing.

(a) What is the frequency from A as heard by the listener?

(b) What is the frequency from B as heard by the listener?

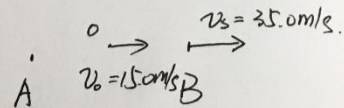
(c) What is the beat frequency detected by the listener?

Week 4

1. Two train whistles, A and B , each have a frequency of 392 Hz. A is stationary and B is moving toward the right (away from A) at a speed of 35.0m/s. A listener is between the two whistles and is moving toward the right with a speed of 15.0m/s (Figure below). No wind is blowing.

(a) What is the frequency from A as heard by the listener?

$$\begin{aligned} f' &= \frac{u - v_o}{u} \cdot f_0 \\ &= \frac{330 - 15}{330} \cdot 392 \text{ (Hz)} \\ &= 374.2 \text{ (Hz)} \end{aligned}$$



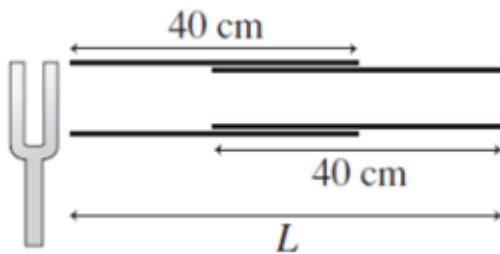
(b) What is the frequency from B as heard by the listener?

$$\begin{aligned} f'' &= \frac{u + v_o}{u + v_s} \cdot f_0 \\ &= \frac{330 + 15}{330 + 35} \cdot 392 \text{ (Hz)} \\ &= 370.5 \text{ (Hz)} \end{aligned}$$

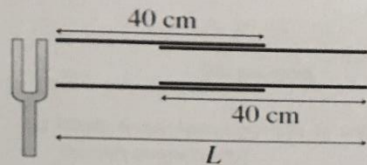
(c) What is the beat frequency detected by the listener?

$$\text{拍频} = |f'' - f'| = 3.7 \text{ (Hz)}$$

2. The 40-cm-long tube of Figure below has a 40-cm-long insert(插入物) that can be pulled in and out. A vibrating tuning fork(振动音叉) is held next to the tube. As the insert is slowly pulled out, the sound from the tuning fork creates standing wave in the tube when the total length L is 42.5cm, 56.7cm and 70.9cm. What is the frequency of tuning fork? Assume the speed of sound is 343 m/s.



2. The 40-cm-long tube of Figure below has a 40-cm-long insert that can be pulled in and out. A vibrating tuning fork is held next to the tube. As the insert is slowly pulled out, the sound from the tuning fork creates standing wave in the tube when the total length L is 42.5cm, 56.7cm and 70.9cm. What is the frequency of tuning fork? Assume the speed of sound is 343 m/s.



$$L = n \cdot \frac{u}{2\Delta} \Rightarrow \begin{cases} L_1 = \frac{n \cdot u}{2\Delta} \\ L_2 = \frac{(n+1)u}{2\Delta} \\ L_3 = \frac{(n+2)u}{2\Delta} \end{cases}$$

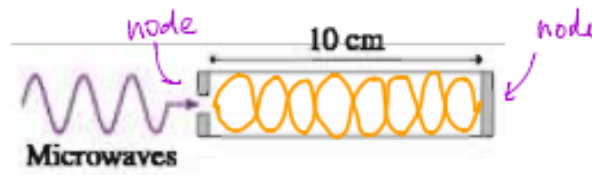
$$\therefore \frac{(n+1) \times 343}{2 \times 56.7 \times 10^{-2}} = \frac{(n+2) \times 343}{2 \times 70.9 \times 10^{-2}} \Rightarrow (n+1) \times 70.9 = (n+2) \times 56.7$$

$$n = \frac{41.5}{14.2} \approx 3$$

$$L_1 = \frac{3 \times u}{2\Delta} \Rightarrow \Delta = \frac{3 \times 343}{2 \times 42.5 \times 10^{-2}}$$

$$= \boxed{1210.6 \text{ (Hz)}}$$

3. The microwave(微波) generator(产生器) in the figure below can produce microwaves at any frequency between 10 GHz and 20 GHz. The microwaves are aimed(对准), through a small hole, into a “microwave cavity(腔)” that consists of a 10-cm-long cylinder(圆柱) with reflective ends.



(a) Which frequencies (between 10 and 20 GHz) will create standing waves in the microwave cavity?

With nodes at both ends, $\lambda_n = \frac{2L}{m}$ $f_1 = \frac{c}{2L} = \frac{3 \cdot 10^8}{2 \cdot 0.1} = 1.5 \text{ GHz}$

$$\rightarrow f_m = \frac{c}{\lambda_m} = \left(\frac{c}{2L} \right) \cdot m$$

m	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$f_m \text{ (GHz)}$	1.5	3	4.5	6	7.5	9	10.5	12	13.5	15	16.5	18	19.5	21

These are in the range 10-20 GHz.

(b) For which of these frequencies is the cavity midpoint an anti-node?

m is the number of anti-nodes.

There will be an anti-node in

the middle if m is odd: $\rightarrow 10.5, 13.5, 16.5, 19.5 \text{ GHz}$

Week 5

essentially(本质上地), film(薄膜)

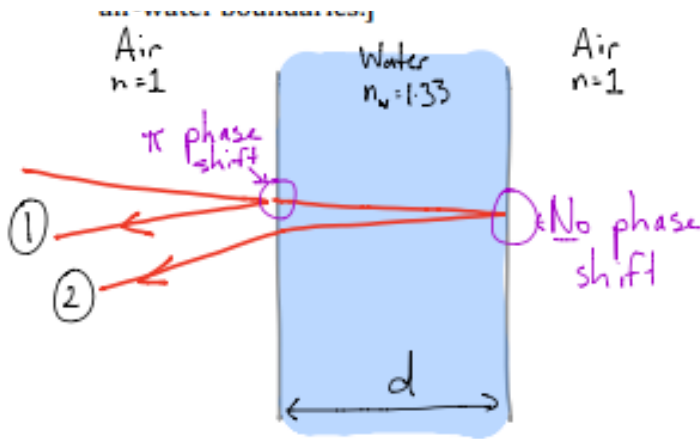
constructive interference(相干加强), destructive interference(相干减弱)

A soap bubble (肥皂泡) is essentially a very thin film of water ($n = 1.33$) surrounded by air. The colors that you see in soap bubbles are produced by interference (干涉).

(a) Derive an expression (求一个公式) for the wavelengths λ_c for which constructive interference causes a strong reflection (强反射) from a soap bubble of thickness d . [HINT: think about the reflection phase shifts at the air-water boundaries.]

(b) What visible wavelengths of light are strongly reflected from a 390-nm-thick soap bubble? What color would such a soap bubble appear to be? [Look at Figure 20.22 on page 17 of the Traveling Waves chapter.]

1.



When rays ① and ② interfere, the phase difference is

$$\Delta\phi = \frac{2\pi}{\lambda_w}(2d) + \pi$$

$\uparrow$ Δr in water $\uparrow$ first reflection

$$= \frac{4\pi n_w d}{\lambda} + \pi$$

For constructive interference,

$$\Delta\phi = m \cdot 2\pi$$

$$\frac{4\pi n_w d}{\lambda} + \pi = m \cdot 2\pi \rightarrow \frac{4\pi n_w d}{\lambda} = \left(m - \frac{1}{2}\right) \cdot 2\pi \rightarrow \lambda_m = \frac{2n_w d}{\left(m - \frac{1}{2}\right)}$$

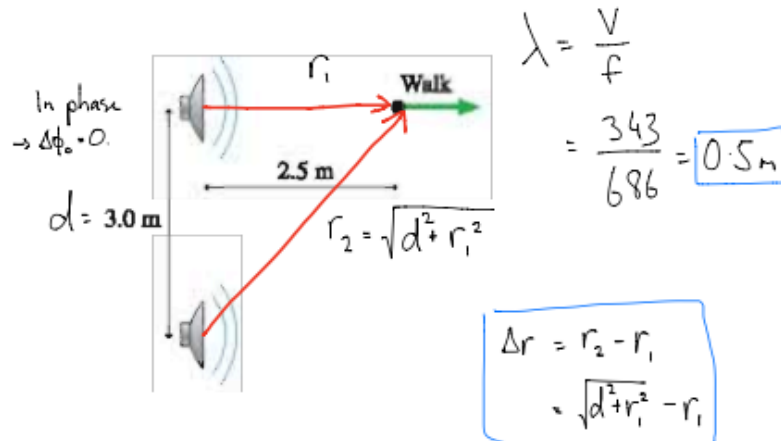
$$d = 390 \text{ nm}, n_w = 1.33 \rightarrow \lambda_m = \frac{2 \times 1.33 \times 390}{(m - \frac{1}{2})} = \frac{1037}{(m - \frac{1}{2})} \text{ nm}$$

m	1	2	3	4	...
λ_m (nm)	2074	692	415	296	...

Only these two
are in the visible
range.

Red + Blue $\rightarrow$ Magenta

2. You are standing 2.5 m directly in front of one of the two loud speakers (音响) shown in the figure below. They are 3.0 m apart and both are playing a 686 Hz tone, in phase (同相位). As you begin to walk directly away from the speaker, at what distances from the speaker do you hear a minimum sound intensity (强度)? Assume the speed of sound in air is 343 m/s.



Condition for destructive interference:

$$\Delta\phi = \frac{2\pi\Delta r}{\lambda} + \Delta\phi_0 = (m + \frac{1}{2}) \cdot 2\pi$$

$$\frac{2\pi\Delta r}{\lambda} = (m + \frac{1}{2}) \cdot 2\pi$$

$$\Delta r = (m + \frac{1}{2}) \lambda \rightarrow \sqrt{d^2 + r_1^2} - r_1 = (m + \frac{1}{2}) \lambda$$

Δr gets smaller as r_1 gets bigger.

Solve for r_1 :

$$\left[\sqrt{d^2 + r_1^2} \right]^2 = \left[(m + \frac{1}{2}) \lambda + r_1 \right]^2$$

$$d^2 + r_1^2 = (m + \frac{1}{2})^2 \lambda^2 + 2(m + \frac{1}{2}) \lambda r_1 + r_1^2$$

$$2(m + \frac{1}{2}) \lambda r_1 = d^2 - (m + \frac{1}{2})^2 \lambda^2 \rightarrow r_1 = \frac{d^2 - (m + \frac{1}{2})^2 \lambda^2}{2(m + \frac{1}{2}) \lambda}$$

Initially, $\Delta r = \sqrt{3^2 + 2.5^2} - 2.5$
 $= 1.4 \text{ m} = 2.8\lambda$

As I walk away, r_1 increases,
 and Δr decreases.

There will be destructive
 interference when

$\Delta r = 2.5\lambda, 1.5\lambda$ and 0.5λ

i.e. $(m + \frac{1}{2}) = 2.5, 1.5$ and 0.5

$\rightarrow r_1 = 3.0 \text{ m}, 5.6 \text{ m}$ and 17.9 m

$$r_1 = \frac{d^2}{2(m + \frac{1}{2})\lambda} - \frac{1}{2}(m + \frac{1}{2})\lambda$$

$$= \frac{3^2}{2 \cdot 0.5(m + \frac{1}{2})} - \frac{1}{2}(0.5)(m + \frac{1}{2})$$

$$= \left[\frac{9}{(m + \frac{1}{2})} - 0.25(m + \frac{1}{2}) \right] \text{ meters}$$

3. Standing waves on a wire are described by $y(x,t) = (A_{SW} \sin kx) \sin \omega t$, with $A_{SW} = 2.50 \text{ mm}$, $\omega = 942 \text{ rad/s}$ and $k = 0.75\pi \text{ rad/m}$. The left end of the wire is at $x=0$. At what distances from the left end are (a) the nodes of the standing wave and (b) the antinodes of the standing wave?

3. Standing waves on a wire are described by $y(x,t) = (A_{SW} \sin kx) \sin \omega t$, with $A_{SW} = 2.50 \text{ mm}$, $\omega = 942 \text{ rad/s}$ and $k = 0.75\pi \text{ rad/m}$. The left end of the wire is at $x=0$. At what distances from the left end are (a) the nodes of the standing wave and (b) the antinodes of the standing wave?

(a) $|\sin kx| = 0 \Rightarrow kx = n\pi \Rightarrow 0.75\pi x = n\pi \Rightarrow x = \left[\frac{4n}{3} \right] (\text{m})$

$x = \frac{4}{3}, \frac{8}{3}, \frac{12}{3}, \dots (\text{m})$

(b) $|\sin kx| = 1 \Rightarrow kx = (2n+1) \cdot \frac{\pi}{2} \Rightarrow 0.75\pi x = (2n+1) \cdot \frac{\pi}{2} \Rightarrow x = \left[\frac{4n+2}{3} \right] (\text{m})$

$x = \frac{2}{3}, \frac{6}{3}, \frac{10}{3}, \dots (\text{m})$

Week6

Week 6

1. A single vertical slit 0.01 mm wide is illuminated by red light of wavelength 700 nm.

(a) About how wide is the bright stripe on a screen 2 meters away?

暗纹: $b \sin \theta = k\lambda$ $\left\{ \begin{array}{l} \Rightarrow \sin \theta = \pm \frac{\lambda}{b} = \pm \frac{700 \times 10^{-9} \text{ m}}{0.01 \times 10^{-3} \text{ m}} = \pm 7 \times 10^{-2} \text{ rad.} \\ k = \pm 1 \end{array} \right.$

$\sin \theta \doteq \frac{x}{f}$

$\Rightarrow x_{\pm 1} = \pm \frac{\lambda f}{b} = \pm 7 \times 10^{-2} \text{ rad} \times 2 \text{ m} = \pm 0.14 \text{ m} \Rightarrow \Delta x = 0.28 \text{ m}$

(b) Does the stripe get wider or narrower if you make the slit narrower?

$\Delta x = \frac{2\lambda f}{b}$, $b \uparrow$ $\Delta x \downarrow$.

$b \downarrow$ $\Delta x \uparrow$.

(c) Does the stripe get wider or narrower if you use green light?

$\Delta x = \frac{2\lambda f}{b}$.

$\lambda \downarrow$ $\Delta x \downarrow$.

2. A ray of violet light (wavelength 400 nm) hits perpendicular to a transmission diffraction grating that has 10,000 lines per centimeter. At what angles to the perpendicular are there bright violet rays, in addition to zero degrees?

$d = \frac{1 \text{ cm}}{1 \times 10^4} = 1 \times 10^{-6} \text{ m}.$

$d \sin \theta = \pm k\lambda \Rightarrow \sin \theta = \frac{\pm k\lambda}{d} = \frac{\pm k \times 400 \times 10^{-9} \text{ m}}{1 \times 10^{-6} \text{ m}} \text{ rad}$

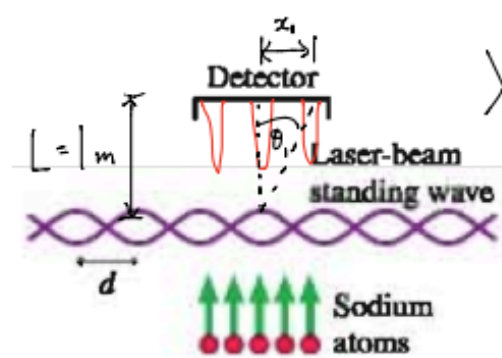
$= \pm 0.4k \text{ rad}$

$\theta = \pm \arcsin(0.4k) \quad (k = 0, 1, 2)$

$\theta = 0^\circ, \pm 23.6^\circ, \pm 53.2^\circ$

3. When we studied diffraction gratings, you learned that light with wavelength λ is diffracted by a piece of matter (the grating) with a periodic(周期性的) structure (many slits with spacing d). Experiments done in the 1990s showed that the roles of light and matter can be reversed; matter with a de Broglie wavelength λ can be diffracted by light with a periodic structure.

The figure below shows a beam(束) of sodium(Na, 钠) atoms (kg) all traveling with a uniform speed of 50 m/s. The atomic beam(原子束) crosses a laser beam(激光光束) standing wave with a wavelength of 600nm. Assuming that the diffraction works just like a diffraction grating, how far will the first-order diffracted atoms be deflected(使偏移) sideways on a detector 1.0 m behind the laser beam?



$$\lambda_{Na} = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(3.84 \times 10^{-26})(50 \text{ m/s})} = 3.45 \times 10^{-10} \text{ m}$$

From the figure, $d = \frac{\lambda_{\text{laser}}}{2} = 300 \text{ nm}$

Constructive interference:

$$\Delta r = d \sin \theta = m \lambda$$

$$\boxed{m=1} \quad \theta_1 = \sin^{-1}\left(\frac{\lambda}{d}\right) \approx \frac{\lambda}{d} \quad (\text{small angle})$$

$$= \frac{3.45 \times 10^{-10}}{300 \times 10^{-9}} = 1.15 \times 10^{-3} \text{ rad}$$

$$x_1 \approx L \theta_1 = (1.0)(1.15 \times 10^{-3}) = 1.15 \times 10^{-3} \text{ m} \\ = \boxed{1.15 \text{ mm}}$$

4. A diffraction grating with 600 lines/mm is illuminated with light of wavelength 500 nm. A very wide viewing screen is 2.0 m behind the grating.

(a) What is the distance between the two $m = 1$ bright fringes(边缘)?

(b) How many bright fringes can be seen on the screen?

4. A diffraction grating with 600 lines/mm is illuminated with light of wavelength 500 nm. A very wide viewing screen is 2.0 m behind the grating.

(a) What is the distance between the two $m = 1$ bright fringes?

$$d = \frac{1 \times 10^{-3}}{600} = \frac{1}{6} \times 10^{-5} \text{ m}$$

$$d \cdot \sin \theta = k\lambda \quad \left. \begin{array}{l} \\ k = \pm 1 \end{array} \right\} \Rightarrow \sin \theta = \pm \frac{\lambda}{d} = \pm \frac{500 \times 10^{-9}}{\frac{1}{6} \times 10^{-5}} = \pm 0.3 \Rightarrow \theta = \pm 17.5^\circ$$

$$\tan \theta = \frac{x}{f} \Rightarrow x = \pm \tan \theta \cdot f = \pm \tan 17.5^\circ \times 2.0 = \pm 0.63 \text{ m}$$

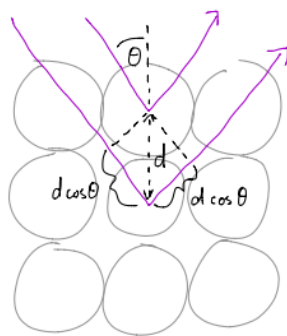
$$\Delta x = 0.63 \text{ m} \times 2 = \boxed{1.26 \text{ m}}$$

(b) How many bright fringes can be seen on the screen?

$$d \cdot \sin \theta = k\lambda \quad \left. \begin{array}{l} \\ \theta = \frac{\pi}{2} \end{array} \right\} \Rightarrow k_m = \frac{d}{\lambda} = \frac{\frac{1}{6} \times 10^{-5} \text{ m}}{500 \times 10^{-9} \text{ m}} = \frac{10}{3} \Rightarrow k_m = \pm 3$$

$\therefore$ 可见 $0, \pm 1, \pm 2, \pm 3$ 共 7 条.

5. X-rays with a wavelength of 0.0700 nm diffract from a crystal. Two adjacent angles of x-ray diffraction are 45.6° and 21.0° . What is the distance between the atomic planes responsible for(由于) the diffraction?



Condition for bright reflection:

$$\Delta r = 2d \cos \theta_m = m\lambda \quad (m=1,2,\dots)$$

$$\text{Let } \theta_m = 45.6^\circ, \theta_{m+1} = 21.0^\circ$$

$$\frac{m+1}{m} = \frac{\cos \theta_{m+1}}{\cos \theta_m} = \frac{\cos 21.0^\circ}{\cos 45.6^\circ} = 1.33 = \frac{4}{3}$$

$$\therefore m = 3, m+1 = 4$$

$$d = \frac{m\lambda}{2 \cos \theta_m} = \frac{3 \times 0.07 \times 10^{-9}}{2 \cos 45.6^\circ} = 1.5 \times 10^{-10} \text{ m} = \boxed{1.5 \text{ \AA}}$$

θ_3

Week 7

1. If a photon(光子) of wavelength 0.04250 nm strikes a free electron(自由电子) and is scattered(散射) at an angle of 35.0° from its original direction, find

- (a) the change in the wavelength of this photon;
- (b) the wavelength of the scattered light;
- (c) the change in energy of the photon (is it a loss or a gain?);
- (d) the energy gained by the electron.

2. (a) A proton is moving at a speed much slower than the speed of light. It has kinetic energy(动能) K_1 and momentum(动量) p_1 . If the momentum of the proton is doubled, so $p_2=2p_1$, how is its new kinetic energy K_2 related to K_1 .

Week 7

1. If a photon of wavelength 0.04250 nm strikes a free electron and is scattered at an angle of 35.0° from its original direction, find

(a) the change in the wavelength of this photon;

$$\Delta\lambda = \frac{h}{m_0c} \cdot (1 - \cos\theta) = 2.43 \times 10^{-12} \text{ m} (1 - \cos 35.0^\circ)$$

$$\boxed{= -4.39 \times 10^{-13} \text{ m}}$$

(b) the wavelength of the scattered light;

$$\lambda = \lambda_0 + \Delta\lambda = (4.250 \times 10^{-11} + 4.39 \times 10^{-13}) \text{ m}$$

$$\boxed{= 4.2939 \times 10^{-11} \text{ m}}$$

(c) the change in energy of the photon (is it a loss or a gain?);

$$\Delta E_{\text{change}} = \frac{hc}{\lambda_0} - \frac{hc}{\lambda} = \frac{h \times 3 \times 10^8 \text{ m/s}}{4.250 \times 10^{-11} \text{ m}} - \frac{h \times 10^8 \times 3 \text{ m/s}}{4.2939 \times 10^{-11} \text{ m}}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{4.25 \times 10^{-11}} - \frac{1}{4.2939 \times 10^{-11}} \bigg) = 4.78 \times 10^{-17} \text{ J}$$

lost

(d) the energy gained by the electron.

$$\Delta E_{\text{electron}} = \Delta E_{\text{change}}$$

$$\boxed{= 4.78 \times 10^{-17} \text{ (J)}}$$

2. (a) A proton is moving at a speed much slower than the speed of light. It has kinetic energy K_1 and momentum p_1 . If the momentum of the proton is doubled, so $p_2=2p_1$, how is its new kinetic energy K_2 related to K_1 .

$$K = \frac{p^2}{2m} \quad \left. \begin{array}{l} p_2 = 2p_1 \\ \end{array} \right\} \Rightarrow \boxed{K_2 = 4K_1}$$

(b) A photon with energy E_1 has momentum p_1 . If another photon has momentum p_2 that is twice p_1 , how is the energy E_2 of the second photon related to E_1 ?

3. A photon of green light has a wavelength of 520 nm. Find the photon's frequency, magnitude(大小) of momentum, and energy. Express the energy in both joules(焦耳) and electron volts(电子伏特).

(b) A photon with energy E_1 has momentum p_1 . If another photon has momentum p_2 that is twice p_1 , how is the energy E_2 of the second photon related to E_1 ?

$$\left. \begin{array}{l} E = \frac{hc}{\lambda} \\ p = \frac{h}{\lambda} \end{array} \right\} \Rightarrow E = pc \Rightarrow \boxed{E_2 = 2E_1}$$

$p_2 = 2p_1$

3. A photon of green light has a wavelength of 520 nm. Find the photon's frequency, magnitude of momentum, and energy. Express the energy in both joules and electron volts.

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{520 \times 10^{-9} \text{ m}} = \boxed{5.8 \times 10^{14} \text{ (Hz)}}$$

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{5.2 \times 10^{-7}} \text{ (kg m/s)}$$

$$= \boxed{1.275 \times 10^{-27} \text{ (kg m/s)}}$$

$$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{5.2 \times 10^{-7}} \text{ (J)} = \boxed{3.825 \times 10^{-19} \text{ (J)}}$$

$$= \frac{3.825 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ (eV)} = \boxed{2.391 \text{ (eV)}}$$